# Python Programming

Sample educational document for testing a PDF question-answering chatbot.

## 1. Variables and types

A variable refers to a value. Common Python types include int for integers, float for decimal numbers, str for text, bool for truth values, list for mutable sequences, and tuple for immutable sequences.

## 2. Functions

A function is a reusable block of code defined with def. Parameters receive input values, and return sends a result back to the caller.

## 3. Lists

A list is an ordered, mutable collection. Indexing accesses an element, slicing selects a range, append adds an element at the end, and len returns the number of elements.

## 4. Dictionaries

A dictionary stores key-value pairs. Keys are used to retrieve corresponding values. Dictionaries are useful for representing structured information such as a student's name and marks.

## 5. Exceptions

Exceptions represent errors or unusual conditions during execution. A try block contains code that may fail, and except blocks handle specified exception types.

*Note: This is a short synthetic sample created for software testing, not an official textbook.*